

## Pipeline Internals

# Pipeline Internals I

This section documents the artefacts produced by the prose-review run.

flowchart TB

P[projects/templates/template\_prose\_project/]

P --> M[manuscript/]

P --> OUT[output/<br/>tracked public artifacts · regenerate]

M --> M\_CFG[config.yaml<br/>single source of truth]

M --> M\_PRE[preamble.md<br/>LaTeX preamble]

M --> M\_SEC[00\_abstract → 04\_conclusion · <br/>05\_pipeline]

M --> M\_BIB[references.bib<br/>read-only · validated by]

OUT --> O\_REP[manuscript\_report.json<br/>ManuscriptReport]

OUT --> O\_CHK[checks.json<br/>list of CheckResult]

OUT --> O\_RR[review\_report.md<br/>final markdown report]

OUT --> O\_DATA[data/manuscript\_variables.json]

## Pipeline Internals II

```
OUT --> O_FIG[figures/<br/>section_word_counts.png ·<br>  
OUT --> O_RS[run_summary.json]
```

```
classDef d fill:#0f172a,stroke:#0f172a,color:#fff  
classDef src fill:#1e3a8a,stroke:#0f172a,color:#fff  
classDef gen fill:#0f766e,stroke:#0f172a,color:#fff  
class P,M,OUT d  
class M_CFG,M_PRE,M_SEC,M_BIB src  
class O_REP,O_CHK,O_RR,O_DATA,O_FIG,O_RS gen
```

# Data structures I

```
classDiagram
class ManuscriptReport {
    +list~FileReport~ files
    +int total_words
    +int total_sentences
    +int total_paragraphs
    +float avg_flesch_reading_ease
    +float avg_flesch_kincaid_grade
    +float avg_gunning_fog
    +list~str~ citation_keys
}

class FileReport {
    +str name
    +ProseMetrics metrics
    +StructureReport structure
    +QualityReport quality
```

## Data structures II

```
}
```

```
class ProseMetrics {  
    +int word_count  
    +int sentence_count  
    +int paragraph_count  
    +float flesch_reading_ease  
    +float flesch_kincaid_grade  
    +float gunning_fog  
}
```

```
class StructureReport {  
    +list~Heading~ headings  
    +list~Section~ sections  
    +int max_depth  
    +bool has_h1  
    +bool has_skipped_level
```

## Data structures III

```
}
```

```
class QualityReport {  
    +int passive_count  
    +int hedge_count  
    +int citation_count  
    +int long_sentence_count  
    +float citation_density_per_1000  
}
```

```
class CheckResult {  
    +str name  
    +bool passed  
    +str message  
}
```

```
class ProseRunArtifacts {
```

## Data structures IV

```
+ManuscriptReport manuscript_report  
+list~CheckResult~ checks  
+bool all_passed  
}
```

```
ManuscriptReport --> "*" FileReport  
FileReport --> ProseMetrics  
FileReport --> StructureReport  
FileReport --> QualityReport  
ProseRunArtifacts --> ManuscriptReport  
ProseRunArtifacts --> "*" CheckResult
```